Naomi Gluck Schlanger

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Education

Yale University | Physics PhD Candidate — Aug. 2021 – Present
Thesis Advisors: Prof. Earl Bellinger, Prof. Daisuke Nagai — *New Haven, CT*

Stony Brook University — Aug. 2017 – May 2021
Bachelor of Science in Physics — *Stony Brook, NY*
Bachelor of Science in Astronomy and Planetary Sciences — *Stony Brook, NY*
Minor in Music — *Stony Brook, NY*

Publications

Refereed Journal Articles: Published

- **Gluck, N.**, Bellinger, E. P., Liang, L., Farag, E., Saunders, N., Kelici, S., Lindsay, C. J. *Benchmarking Machine Learning Emulators of Stellar Evolution for Precision Asteroseismology*, 2026, ApJ 1010, 1
- Adams, K., Oppenheimer, B., **Gluck, N.**, Ogle, C., Ho, M., & Nagai, D. *CNN-Based Inference of Gaseous Halo Properties from Synthetic X-ray and 21-cm HI Observations*, 2026, submitted to OJA
- Ogle, C., Adams, K., Oppenheimer, B., Ho, M., **Gluck, N.**, & Nagai, D. *The Deep Learning Halo Definer: A Multimodal Framework for Halo Mass and Gas Fraction Inference on Galaxy Groups and Clusters*, 2026, submitted to OJA
- **Gluck, N.**, Oppenheimer B. D., Nagai, D., Villaescusa-Navarro V., & Anglés-Alcázar, D., *An Observationally Driven Approach for Probing the Circum-Galactic Medium with Convolutional Neural Networks*, 2023, MNRAS, 527, 10038
- Gofman, R., A., **Gluck, N.**, & Soker, N., *Enhanced mass-loss rate evolution of stars with mass greater than 18 $M_\odot$, and missing optically-observed type II supernovae*, 2020, MNRAS 494, 5230

Refereed Journal Articles: in-prep

- **Gluck, N.**, Bellinger, E. P., Liang, L., Farag, E., Saunders, N., Kelici, S., Lindsay, C. J. *`SEISTRON`: Stellar Evolution and Interior STructures with vaRiability and Optimized Neural networks*, to be submitted to ApJ
- **Gluck, N.**, Lau, E. T., Nagai, D., *Differentiable Baryon Pasting Model with JAX: Impact of Halo Formation History on the thermal SZ Angular Power Spectra*, to be submitted to ApJ
- Warburton, I., **Gluck, N.**, Nagai, D., Ntampaka, M., Aung, H., Bose, S., *Cosmology Dependence of the Universal Mass Accretion Rate of Dark Matter Halos*, to be submitted to ApJ

Galactic Atmosphere Articles

- Singh, P., Nagai, D., Oppenheimer, B. D., Lau, E., **Gluck, N.**, & Medlock, I.. Galactic Gaseous Halos: Mini-Clusters Disrupted by Feedback, 2022, Galactic Atmospheres.
- Oppenheimer, B. D., Nagai, D., Lau, E., Singh, P., Contreras, A. B., **Gluck, N.**, Jones, J. D., Medlock, I., & Villaescusa-Navarro, F.. A Multi-Wavelength, Multi-Model Exploration of How Feedback Disrupts Gaseous Atmospheres, 2022, Galactic Atmospheres.

Awards

D. Allan Bromley Graduate Fellowship in Physics (2025-2026), Yale University, Awarded $5,000

Dean's Fund for Symposia and Colloquia, Yale University
Physics x Data Science Seminar Series 2025-2026 – Awarded $800
Physics x Data Science Seminar Series 2024-2025 – Awarded $1,000

Associate in Teaching Fellowship, Yale University, Awarded $11,070
PHYS 378 (Introduction to Scientific Computing and Data Science – Fall 2024)

Presentations

Astronomy x Data Science Seminar, 10/2026, 07/2026, Yale University (talk)

Yale/CeNAM Nuclear Asteroseismology Workshop, 07/2026, Yale University (talk)

FDS Workshop: AI for Scientific Discovery, 05/2026, Yale University (talk)

European Southern Observatory - AI Forum, 04/2025 (talk)

European Astronomical Society - Annual Meeting, 07/2024 (poster)

ML-IAP/CCA Debating the Potential of Machine Learning in Astronomical Surveys, 11/2023 (poster)

Astronomy x Data Science Seminar, 09/2023, Yale University (talk)

APS April Meeting 2023, Minneapolis, MN (poster)

KITP Workshop, 02/2023, CCA at Flatiron Institute (talk)

CAMELS Workshop, 12/2022, CCA at Flatiron Institute (talk)

GAINS Conference, 04/2022, Yale University (talk)

IACS Seminar, 04/2021, Stony Brook University (invited talk, with Prof. Douglas Swesty)

Research Experience

Graduate Research September 2021 – Present

Yale University, Prof. Earl Bellinger and Prof. Daisuke Nagai *New Haven, CT*

- Ph.D Thesis: "Stellar Evolution to Galaxy Evolution: Soldering the Gap"
- SEISTRON (Stellar Evolution and Internal Structures from vaRiability with Optimized Neural networks): A machine-learning based open-science package for stellar phenomenology.
- Machine Learning with CAMELS (Cosmology and Astrophysics with MachinE Learning Simulations): Using convolutional neural networks to infer properties of the circum-galactic medium with idealized and observationally limited 2D maps of HI and X-ray.
- Baryon Pasting (BP) Collaboration: Building a differentiable gas model with mass accretion history and cosmological dependence as an update to the current BP pipeline. Developing a machine-learning BP approach using multi-fidelity models.

Argonne National Laboratory, Dr. Andrew Hearin *Lemont, IL*

- May 2022 – August 2022: Halo-Galaxy Connection Forward Modeling: Implemented new differentiable pipeline for the orbital evolutionary history galactic halos and substructure into the SatGen galaxy evolution and subhalo tracking model.

Undergraduate Research 2019 – 2021

Stony Brook University *Stony Brook, NY*

- Physics Senior Thesis, Advised by Prof. Alan Calder: Uncertainty Quantification for the evolution of solar-mass stars to white dwarfs following the MESA (Modules for Experiments in Stellar Astrophysics) open source code to determine the validity and bounds of wind parameters.
- Astronomy Senior Thesis, Advised by Prof. Fredrick Walter: Data analysis of Nova V1047 using archival spectroscopic data from Stony Brook/SMARTS to perform a spectral time analysis on two different events.

Technion Institute of Technology, Prof. Noam Soker *Haifa, Israel*

- June 2019 – April 2020: Simulated the evolution of observationally obscured Type-II Core-Collapse Supernovae using MESA open source code with metal abundance and instability calculations.

Manuscript Reviewer

Monthly Notices of the Royal Astronomical Society (MNRAS)

Publications of the Astronomical Society of the Pacific (PASP) | Star Reviewer Status

Leadership Roles

Collaboration Memberships: CAMELS, Baryon Pasting, CMB-S4, Rubin LSST

Physics x Data Science Seminar Series (2024 – 2026): Leader and co-organizer of bi-weekly seminar series, aiming to bridge the gap between the Physics and Statistics & Data Science departments, fostering discussions and cross-departmental, cross-subfield research collaborations.

Interpretable Machine Learning Working Group at Yale (2023 – 2026): Leader and organizer of bi-weekly meetings on the importance of machine learning interpretability in computational research, including seminars/talks from invited external speakers, project updates from group members, and discussions on state-of-the-art analysis techniques.

Stony Brook University Leadership Roles:
Seawolves for Israel | Secretary (2018-19), Vice President (2019-20), President (2020-21)
Hillel Board of Directors | Student Representative (2020 – 2021)
University Orchestra | Principal Oboist (2017 – 2021)

Mentoring of Undergraduate Student Research in Nagai Lab

Tanish Chettiar Yale Physics, Fall 2024 – Spring 2025
Junior Research: *"Glitch Classification with Machine Learning"* (co-mentored with Rhudresh Manoharan, Baylor)

Din-Ammar Tolj Yale Physics, Fall 2023 – Spring 2025
Junior Research: *"Modeling Gas Shape of Dark Matter Halos in TNG-300"*
Summer Research: *"Characterizing Triaxial Dark Matter Halo Shapes with Symbolic Regression"*

Daniel Chang Yale Physics, Fall 2023 – Spring 2024
Senior Thesis: *"Probing CGM Physics with Interpretable Machine Learning"*

Finn Gibson Yale Physics, Fall 2023
Senior Thesis: *"Correlation of Dark Matter, Gas & Stellar Profiles in CAMELS Simulations"*

William Kline Yale Applied Mathematics, Spring 2023
Senior Thesis: *"Modeling Dynamical Friction of Infalling Cluster Galaxies"* (co-mentored with Han Aung)

Teaching and Tutoring

Associate in Teaching Fellowship, Yale University
PHYS 378 (Introduction to Scientific Computing and Data Science – Fall 2024)

Guest Instructor, Yale University
PHYS 378 (Introduction to Scientific Computing and Data Science – Fall 2023)

Graduate Teaching Fellow, Yale University
PHYS 200 (Fundamentals of Physics – Fall 2021)
PHYS 120 (Quantum Physics and Beyond – Spring 2022)
PHYS/ASTR 343 (Gravity, Astrophysics, and Cosmology – Fall 2022, Spring 2025)
PHYS 378 (Introduction to Scientific Computing and Data Science – Spring 2023)

Private STEM Tutoring (2016 – Present): In-person and online in Physics (Regents, Honors, AP Physics 1, AP Physics C), Math (Algebra, Trigonometry, Geometry, Pre-Calculus, AP Calculus AB, AP Calculus BC, Exeter Calculus Programs), Biology, Chemistry, and Earth Science.

Technical Skills

Computational Science: Techniques of parallel computing including parallelization by both threads (OpenMP) and message passing (MPI), job submission with Slurm, and software management with Modules.
Languages: Python, C/C++, Matlab, Fortran, Mathematica
Software Skills: MESA, DS9, CCDSoft, SkyChart, Procreate, LTSpice, Sibelius, MuseScore

Outreach Activities

High School Talks – Physics and Academia, Trial and Error
Syosset High School (3/2023 – Hosted by Jill Johansen)
Notre Dame Academy - All Girl High School (12/2022 – Hosted by Owen Steele)

Graduate Affiliate at Yale, Berkeley College (2022 – 2024): Serving as a mentor and providing undergraduate students with one-on-one advice on navigating their post-graduation journey.

Stony Brook University Leadership Roles:

Seawolves for Israel | Secretary (2018-19), Vice President (2019-20), President (2020-21)
Hillel Board of Directors | Student Representative (2020 – 2021)
University Orchestra | Principle Oboe (2017 – 2021)

Work Experience

BOOST Tutors and Mentors (September 2021–Present): Physics (regents, honors, AP Physics 1, AP Physics C), Math (algebra/trigonometry, geometry, pre-calculus, AP Calculus AB, AP Calculus BC, Exeter Math Programs), and Biology. Preparing students for course exams, the math and science sections of SAT/ACT exams, and AP Exams.

Business Partnership - ANG Designs.co (2020 – Present): Established online custom graphics art company, specifically partnering with Stony Brook University Hillel, SUNY Geneseo Hillel, and the Ohio State Hillel.

StandWithUs Emerson Fellowship (August 2019 – May 2020): Partnered with other student organizations at Stony Brook University to create 12 Israel-related events that impacted approximately 150 students.